

Slowly Varying Critical Transseries

Universal action budgets, convergent Lambert charts,
and an effective-index expansion

Research article prepared for Vladimir Reshetnikov

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Abstract

Consider the countable exponential-feedback equation $U = \beta A(qe^U)$ with nonnegative coefficients $a_j \sim aj^{-1-\alpha}\ell(j)$, where $1 < \alpha < 2$ and ℓ is slowly varying. The ProveIt critical-Hahn manuscript explicitly asks whether its exact-power-tail theory extends to this class. We give a fixed-interior-exponent answer. The critical inverse is described by an asymptotic inverse of $y^\alpha/\ell(y)$; the same normalization controls the large coefficients and the sharp scale of asymptotically sufficient action cutoffs. A compensated stable density supplies a universal coupling-cutoff profile, independent of ℓ after normalization, and a conditional point-process formula identifies the entire cloud of large actions.

The representation question has a genuine obstruction: an explicit positive, slowly varying oscillatory tail yields no leading monomial in any finite iterated power-log scale. For the exact tails $j^{-1-\alpha}(\log j)^m$, $m \geq 1$, this obstruction disappears. We construct an absolutely convergent three-coordinate analytic chart over a W_{-1} core, with explicit first coefficients and a geometric tail bound. We also prove all-order inverse-logarithmic expansions for coefficients and cutoff probabilities. Their first correction is exactly the derivative obtained by replacing α by $\alpha - m/\log b_n$ while holding the normalized coordinates fixed. A positive derivative formula for the cutoff profile turns this into a two-term minimal-budget law.

The stable-limit and regular-variation mechanisms are classical. This article supplies explicit inverse charts, differential correction operators, and action-budget formulas, with proofs connecting them to that classical theory. No claim of historical priority, endpoint uniformity, or Lean formalization is made. The package contains 552 exact algebra checks and separately labelled floating-point diagnostics.

Keywords. Transseries; Lagrange inversion; regular variation; stable local limits; action truncation; Lambert W ; logarithmic asymptotics.

Contents

1	The question being answered and the limits of the answer	3
1.1	Repository provenance	3
1.2	Results and their logical scope	3
1.3	Classical input and novelty discipline	3
2	Model, normalization, and exact coefficient identities	4
3	The critical inverse for an arbitrary slowly varying tail	5
4	A genuine obstruction to power–logarithmic representation	6
5	Stable local limits from a direct Fourier argument	7
6	The universal cutoff profile and the whole large-action cloud	9
6.1	A marked point-process formulation	10
7	Exact logarithmic tails and a convergent Lambert chart	11
7.1	A Lambert core that keeps the leading logarithm exact	12
7.2	A genuine geometric error bound	13
8	All-order logarithmic coefficient and cutoff expansions	13
8.1	The effective-index law	15
8.2	Higher orders are not just an index shift	16
9	Two-term minimal action budgets	16
10	Reproducible calculations and what they actually verify	17
10.1	Exact checks	17
10.2	A logarithmic-tail example	17
10.3	Inverse-chart diagnostics	18
10.4	Numerical method and independent cross-checks	19
11	Consequences for a transseries implementation	19
11.1	Three distinct kinds of representation	19
11.2	A practical coefficient pipeline	19
11.3	A realistic formalization boundary	20
12	Further research questions and concrete targets	20
13	Conclusion	21
A	Normalization and sign audit	22
B	Reproduction instructions	22

1 The question being answered and the limits of the answer

1.1 Repository provenance

The repository was inspected at commit

ebd8344bca77d8352cd745b7df374618a290a029.

The primary predecessor is *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, located under the transseries documentation directory in `Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/` [1]. Its research Question 1, “Slowly varying action tails,” asks for the implicit stable scale, the cutoff profile, an inverse notation when de Bruijn conjugates are necessary, and hypotheses preserving a finite positive support grid. That question is the direct target of this article.

The predecessor already treats exact power tails, a bounded coupling window, and the conditional-Poisson interpretation of action truncation. These are not claimed as new here. The repository index also records upper-endpoint articles involving cubic tails and logarithmic modifications [2]. Our exponent stays in $(1, 2)$; neither an $\alpha \rightarrow 2$ limit nor the Gaussian endpoint is being repeated or asserted. The repository index warns that its recent research arrivals are unmerged and not formally verified. We therefore prove the mathematical statements needed below rather than use those arrivals as unexamined analytic axioms.

Editorial note (ProveIt, 2026-09-29). The logarithmic-endpoint package (`Logarithmic_Critical_Endpoint_Lambert_Charts/` [8], in the tree at the inspected snapshot) treats the $\alpha = 2$ analogue of the tails of section 7, $a_j \sim aj^{-3}(\log j)^r$, $r \geq -1$, at leading order (its `thm:weightedchart`, `thm:weightedbudget`); its Lambert core has the form of (7.7) at $(\alpha, m) = (2, 1)$, where the logarithm comes from the pole of $\Gamma(-\alpha)$. The marginal package (`Marginal_Critical_Transseries_Action_Budgets/` [9], `thm:min-budget`) gives the $\alpha = 2$ two-term minimal budget; the later confluent and stable-Gaussian packages [10, 11] treat the crossover $\alpha \rightarrow 2$ for exact tails, where the cutoff profile tends to $\exp(-s^{-\alpha}/\alpha)$. None of them treats $1 < \alpha < 2$ with a slowly varying factor.

1.2 Results and their logical scope

For arbitrary positive slowly varying ℓ , we prove a critical real inverse law, a stable local limit, a two-parameter cutoff profile, and the exact criterion $M_n/b_n \rightarrow \infty$ for asymptotically lossless truncation in a bounded critical window. These conclusions require no complex continuation of ℓ . An oscillatory example shows why they cannot automatically be upgraded to a power-logarithmic transseries representation.

For the smaller, analytically controlled class with eventual exact weights $aj^{-1-\alpha}(\log j)^m$, we prove much more: a convergent local inverse chart, all fixed orders of a coefficient expansion, the corresponding cutoff expansion, and an asymptotic minimal-budget formula. The coefficient expansion is asymptotic, not asserted to converge; the local inverse chart really does converge. Confusing those two statements would erase the main analytic distinction of the paper.

1.3 Classical input and novelty discipline

Regular variation, asymptotic inversion, and de Bruijn conjugacy are part of the classical theory developed in [3]. The relation between generating functions, conditioned allocations, stable limits, and largest components is classical as well; Janson’s survey is a detailed reference [4].

Our probability proof is a direct Fourier proof for this specific compound-Poisson model, not a claim to have invented a stable local limit theorem. The convergent special-function expansion used for exact logarithmic tails is the parameter-differentiated polylogarithm expansion in [5]; the Lambert branch is standard [6]. Transseries terminology and the difference between formal operations and analytic realization are discussed in [7].

The principal synthesis developed here is the following: the same slowly varying normalization controls inverse singularities and action budgets; logarithmic tails give an explicit convergent inverse atlas; and logarithmic perturbations of the Lévy measure induce a common differential correction on the density, the retained fraction, and its quantiles. These are results proved in this article, not independently certified claims of publication priority. The literature comparison is targeted rather than exhaustive.

2 Model, normalization, and exact coefficient identities

Fix $1 < \alpha < 2$, $a > 0$, and a sequence $(a_j)_{j \geq 1}$ satisfying

$$a_j \geq 0, \quad a_1 > 0, \quad a_j \sim a j^{-1-\alpha} \ell(j), \quad (2.1)$$

where ℓ is eventually positive, measurable, locally bounded, and slowly varying at infinity:

$$\frac{\ell(cy)}{\ell(y)} \longrightarrow 1 \quad \text{for every fixed } c > 0.$$

Values of ℓ on an initial bounded interval are immaterial. Set

$$A(t) = \sum_{j \geq 1} a_j t^j, \quad d_0 = A(1), \quad d_1 = \sum_{j \geq 1} j a_j, \quad \beta_c = d_1^{-1}, \quad \rho_\beta = e^{-\beta d_0}. \quad (2.2)$$

Both d_0 and d_1 are finite and positive. The second moment is infinite. For $\beta > 0$ define

$$U_\beta(q) = \beta A(q e^{U_\beta(q)}) = \sum_{n \geq 1} u_n(\beta) q^n. \quad (2.3)$$

A formal coefficient recursion and the analytic implicit-function theorem at $(q, U) = (0, 0)$ give the same unique analytic germ. Nonnegativity and $a_1 > 0$ give $u_n(\beta) > 0$ for all $n \geq 1$.

For a positive integer M replace A by $A_M(t) = \sum_{j \leq M} a_j t^j$ and write $u_{n,M}(\beta)$ for the resulting coefficients. This is a literal removal of actions, without renormalizing the remaining weights or changing β .

Proposition 2.1 (Exact coefficient and conditioning identities). *For $n \geq 1$,*

$$u_n(\beta) = \frac{1}{n} [t^n] e^{n\beta A(t)}. \quad (2.4)$$

Let $N_{n,j}$ be independent Poisson variables of means $n\beta a_j$ and $K_n = \sum_{j \geq 1} j N_{n,j}$. Then

$$u_n(\beta) = \frac{\rho_\beta^{-n}}{n} \mathbb{P}(K_n = n), \quad (2.5)$$

$$\frac{u_{n,M}(\beta)}{u_n(\beta)} = \mathbb{P}(N_{n,j} = 0 \text{ for every } j > M \mid K_n = n). \quad (2.6)$$

The ratio is nondecreasing in M , belongs to $[0, 1]$, and equals 1 for $M \geq n$.

Proof. Put $t = qe^{U_\beta}$, so $t = qe^{\beta A(t)}$ and $U_\beta = \beta A(t)$. Lagrange inversion gives

$$[q^n]U_\beta = \frac{1}{n}[t^{n-1}]\beta A'(t)e^{n\beta A(t)} = \frac{1}{n}[t^n]e^{n\beta A(t)}.$$

The total Poisson intensity is $n\beta d_0 < \infty$, so only finitely many $N_{n,j}$ are nonzero almost surely. Its generating function is $\mathbb{E}t^{K_n} = \exp(n\beta(A(t) - d_0))$, proving (2.5). On the no-large-action event, independence leaves $\exp(n\beta(A_M(t) - d_0))$. Comparing its coefficient with the full one proves (2.6). Its remaining properties follow from nested events. $\square$

Definition 2.2 (Critical scale and coupling coordinate). Let $\beta_n \rightarrow \beta_c$ and choose $b_n \rightarrow \infty$ such that

$$n\beta_n a \ell(b_n) b_n^{-\alpha} \rightarrow 1. \quad (2.7)$$

Define

$$\lambda_n = \frac{n(1 - \beta_n d_1)}{b_n}. \quad (2.8)$$

Unless stated otherwise, the coupling coordinate is confined to a fixed compact subset of $\mathbb{R}$.

Such a scale exists by asymptotic inversion of a regularly varying function of positive index. It is unique up to asymptotic equivalence and has index $1/\alpha$ as a function of n . In particular $b_n = o(n)$. Replacing it by an asymptotically equivalent scale changes neither the leading laws nor the criterion for a lossless cutoff. Higher-order laws will require a more precise normalization.

3 The critical inverse for an arbitrary slowly varying tail

Write $t = e^{-x}$ and let $q = te^{-\beta_c A(t)}$. The exact real boundary coordinate and inverse value are

$$\varepsilon = \log(\rho_c/q) = x + \beta_c(A(e^{-x}) - d_0), \quad (3.1)$$

$$U_{\beta_c}(q) = \beta_c d_0 + \varepsilon - x. \quad (3.2)$$

Here $x \downarrow 0$ as $q \uparrow \rho_c$. The derivative of the right side of (3.1) is $1 - \beta_c \sum j a_j e^{-jx} > 0$ for $x > 0$. Thus the positive inverse exists and is unique near the boundary.

Lemma 3.1 (Compensated Abelian limit). *Under (2.1),*

$$A(e^{-x}) - d_0 + d_1 x \sim a\Gamma(-\alpha)x^\alpha \ell(1/x) \quad (x \downarrow 0). \quad (3.3)$$

Proof. Put $h(v) = e^{-v} - 1 + v$. The expression on the left is $\sum a_j h(jx)$. On v in a compact subinterval of $(0, \infty)$, regular variation converts its normalization by $ax^\alpha \ell(1/x)$ into the Riemann sum for $\int h(v)v^{-1-\alpha} dv$. The uniform convergence theorem and Potter bounds for slowly varying functions justify the passage from compact subintervals to the full half-line [3]. For completeness, choose $0 < \eta < \min(2 - \alpha, \alpha - 1)$. Near zero use $h(v) \leq v^2/2$ and the bound $Cv^{-1-\alpha-\eta}$ for the normalized weight; near infinity use $h(v) \leq v$ and $Cv^{-1-\alpha+\eta}$. The two dominating functions are integrable. An initial finite prefix contributes $O(x^2)$, which is $o(x^\alpha \ell(1/x))$. Two integrations by parts give

$$\int_0^\infty h(v)v^{-1-\alpha} dv = \frac{\Gamma(2 - \alpha)}{\alpha(\alpha - 1)} = \Gamma(-\alpha) > 0.$$

All boundary terms vanish by $1 < \alpha < 2$. $\square$

Let $R(y) = y^\alpha / \ell(y)$. We use $R^\leftarrow$ to denote any asymptotic inverse, not necessarily the ordinary inverse of the original possibly nonmonotone representative. One can first replace R by an asymptotically equivalent increasing function.

Theorem 3.2 (Critical inverse and conjugate notation). *Put $C = \beta_c a \Gamma(-\alpha)$. The solution of (3.1) satisfies*

$$x(\varepsilon) \sim \frac{1}{R^\leftarrow(C/\varepsilon)}. \quad (3.4)$$

In particular x is regularly varying at zero with index $1/\alpha$. If $k(y) = \ell(y)^{-1/\alpha}$ and $k^\#$ is a de Bruijn conjugate, then, with $z = (C/\varepsilon)^{1/\alpha}$,

$$x(\varepsilon) \sim \frac{1}{z k^\#(z)}, \quad b_n \sim (n \beta_n a)^{1/\alpha} k^\#((n \beta_n a)^{1/\alpha}). \quad (3.5)$$

Proof. By theorem 3.1, $\varepsilon \sim C/R(1/x)$. Asymptotic inversion of a regularly varying function with positive index proves (3.4). The conjugacy relations are

$$k(y)k^\#(yk(y)) \rightarrow 1, \quad k^\#(y)k(yk^\#(y)) \rightarrow 1.$$

Since $R(y) = (yk(y))^\alpha$, the second relation implies $R(zk^\#(z)) \sim z^\alpha$. This proves the first formula in (3.5), and the same argument applied to (2.7) proves the second. Existence of a conjugate and uniqueness up to asymptotic equivalence are the standard conjugacy theorem of [3]. $\square$

Corollary 3.3 (Reciprocal matching). *At $\beta_n = \beta_c$,*

$$\varepsilon(1/b_n) \sim \frac{\Gamma(-\alpha)}{n}.$$

Thus the inverse singularity scale and the large-coefficient fluctuation scale are reciprocal, with the displayed gamma normalization.

Proof. Substitute $x = 1/b_n$ into theorem 3.1 and use (2.7). $\square$

This theorem is a real asymptotic statement. Slow variation alone does not supply analytic continuation of A across its unit-circle boundary, a complex inverse atlas, or a convergent Hahn expansion. The following counterexample shows that even the leading monomial can fail.

4 A genuine obstruction to power–logarithmic representation

Theorem 4.1 (A slowly varying tail without a power–log leading monomial). *Take, for all sufficiently large j ,*

$$a_j = a j^{-1-\alpha} (2 + \sin(\log \log j)), \quad (4.1)$$

and choose any nonnegative finite prefix with $a_1 > 0$. Its positive critical inverse $x(\varepsilon)$ is not asymptotic to any expression

$$c \varepsilon^r \prod_{h=1}^d (\log_h(1/\varepsilon))^{s_h}, \quad c > 0, \quad r, s_h \in \mathbb{R}, \quad d < \infty, \quad (4.2)$$

where $\log_h$ denotes the h -fold iterated logarithm. Consequently it has no convergent well-based expansion in these monomials whose first nonzero term is asymptotically dominant. In particular no ordinary finite Puiseux chart exists.

Proof. The factor $\ell(y) = 2 + \sin(\log \log y)$ is positive and slowly varying: for fixed $c > 0$, the change in its sine argument is $O(1/\log y)$, and $\ell(y) \geq 1$. By [theorem 3.2](#), any monomial equivalent must have $r = 1/\alpha$. The inequalities $1 \leq \ell \leq 3$ and [theorem 3.1](#) imply that $x(\varepsilon)/\varepsilon^{1/\alpha}$ is bounded above and below by positive constants. In a finite product of iterated logarithmic powers, the first nonzero exponent decides whether the product tends to zero or infinity. Therefore all s_h in (4.2) must vanish.

It remains to disprove convergence to a constant. Let

$$x_k = \exp\{-\exp(2\pi k + \pi/2)\}, \quad y_k = \exp\{-\exp(2\pi k + 3\pi/2)\}.$$

Along these two sequences $\ell(1/x_k) = 3$ and $\ell(1/y_k) = 1$. With $C = \beta_c a \Gamma(-\alpha)$, [theorem 3.1](#) yields

$$\frac{x_k}{\varepsilon(x_k)^{1/\alpha}} \rightarrow (3C)^{-1/\alpha}, \quad \frac{y_k}{\varepsilon(y_k)^{1/\alpha}} \rightarrow C^{-1/\alpha}.$$

The limits differ. A well-based expansion with a dominant leading term would provide the forbidden monomial equivalent. $\square$

Remark 4.2 (What the obstruction does not say). The theorem concerns the specified finite iterated power–log scales and expansions with their usual leading-term meaning. It does not exclude a calculus enlarged by oscillatory slow blocks, and it is not a theorem that all possible generalized transseries fields fail. It does show that “slowly varying” is not a sufficient hypothesis for the repository’s finite positive power–log grid to survive unchanged.

5 Stable local limits from a direct Fourier argument

The normalization throughout is the Lévy measure

$$\nu_\alpha(dx) = x^{-1-\alpha} \mathbf{1}_{x>0} dx. \quad (5.1)$$

Define

$$J_\alpha(t) = \int_0^\infty (e^{itx} - 1 - itx)x^{-1-\alpha} dx = \Gamma(-\alpha)(-it)^\alpha, \quad (5.2)$$

$$g_\alpha(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity + J_\alpha(t)} dt. \quad (5.3)$$

The complex power uses the principal branch. Since $\cos(\pi\alpha/2) < 0$, the Fourier integral converges absolutely and remains so after multiplication by any polynomial in t or $\log|t|$. The corresponding probability law can be constructed directly: compensate the Poisson jumps on $(0, 1]$ and pass to their L^2 limit, using $\int_0^1 x^2 \nu_\alpha(dx) < \infty$; then add the independent, finite-intensity jumps on $(1, \infty)$ and subtract their finite mean. Its characteristic exponent is J_α . The same construction with an upper jump cutoff defines the bounded-jump laws used below. At zero,

$$g_\alpha(0) = \frac{\Gamma(-\alpha)^{-1/\alpha}}{\alpha\Gamma(1-1/\alpha)} = \frac{\Gamma(-\alpha)^{-1/\alpha}}{|\Gamma(-1/\alpha)|}. \quad (5.4)$$

One obtains this directly by integrating $\exp\{-zt^\alpha\}$ on $t > 0$ with $z = -\Gamma(-\alpha)e^{-i\pi\alpha/2}$ and applying the reflection formula.

Lemma 5.1 (Scaled measure convergence and Fourier damping). *Let (2.7) hold and put $\nu_n = n\beta_n \sum_{j \geq 1} a_j \delta_{j/b_n}$. Then*

$$\int (e^{itx} - 1 - itx) \nu_n(dx) \longrightarrow J_\alpha(t) \quad (5.5)$$

uniformly for t in compact sets. If $\phi_n(t)$ is the exponential of the left side, its restriction to $[-\pi b_n, \pi b_n]$ has uniformly integrable tails. More explicitly, for sufficiently small fixed $\eta > 0$ there are $c, C, T > 0$ such that in the small-frequency region, for $T \leq |t| \leq \theta_0 b_n$,

$$|\phi_n(t)| \leq C e^{-c|t|^{\alpha-\eta}}, \quad (5.6)$$

and the remaining fixed-angle region contributes $O(b_n e^{-cn})$. The same estimates hold after retaining only jumps $j \leq M_n$, uniformly when M_n/b_n lies in a compact subset of $(0, \infty)$.

Proof. Vague convergence $\nu_n \rightarrow \nu_\alpha$ follows from regular variation and Riemann sums. For $0 < \eta < \min(2 - \alpha, \alpha - 1)$, Potter bounds give uniformly

$$\int_{(0,r]} x^2 \nu_n(dx) \leq C r^{2-\alpha-\eta} + o(1), \quad \int_{[R,\infty)} x \nu_n(dx) \leq C R^{1-\alpha+\eta}.$$

The initial finite prefix contributes $O(n/b_n^2) = o(1)$. These inequalities and the bounds on the compensated exponential used in [theorem 3.1](#) prove (5.5).

For damping, write $\theta = t/b_n$. The negative logarithm of the modulus is $n\beta_n \sum a_j(1 - \cos(j\theta))$. For small nonzero θ , restrict j to $1/|\theta| \leq j \leq 2/|\theta|$. On this interval $1 - \cos(j\theta)$ is bounded below by a positive constant. The sum is at least $c|\theta|^{\alpha\ell}(1/|\theta|)$. Dividing by the normalization at b_n and applying a lower Potter bound gives $c|t|^{\alpha-\eta}$ when $1/|\theta|$ is sufficiently large. For all remaining angles bounded away from zero, $a_1 > 0$ gives $n\beta_n a_1(1 - \cos \theta) \geq cn$ on $[-\pi, \pi]$.

For a cutoff with $M_n/b_n \geq s_- > 0$, the same block of indices is retained once $|t| \geq 2/s_-$. The finite interval of smaller t does not affect tail integrability. The $j = 1$ term is retained eventually as well. $\square$

Theorem 5.2 (Uniform stable coefficient law). *Under (2.1) and (2.7),*

$$\sup_{k \in \mathbb{Z}} \left| b_n \mathbb{P}(K_n = k) - g_\alpha \left(\frac{k - n\beta_n d_1}{b_n} \right) \right| \rightarrow 0. \quad (5.7)$$

In particular, in every bounded coupling window,

$$u_n(\beta_n) = \frac{\rho_{\beta_n}^{-n}}{n b_n} (g_\alpha(\lambda_n) + o(1)). \quad (5.8)$$

The error is relative $o(1)$ uniformly when λ_n stays in a compact set.

Proof. Fourier inversion on the integer lattice gives

$$b_n \mathbb{P}(K_n = k) = \frac{1}{2\pi} \int_{-\pi b_n}^{\pi b_n} e^{-it(k - n\beta_n d_1)/b_n} \phi_n(t) dt.$$

By [theorem 5.1](#), the integrands without the unit-modulus phase converge in $L^1(\mathbb{R})$ after extension by zero. This proves the supremum statement independently of k . The stable density is strictly positive on $\mathbb{R}$; a direct positivity argument is included in the next section. Its positive minimum on each compact set upgrades the result to relative error there. Equation (2.5) completes the proof. $\square$

The exponential factor $\rho_{\beta_n}^{-n}$ must not be replaced by ρ_c^{-n} merely because $\beta_n \rightarrow \beta_c$. Their ratio can be exponentially large or small on the critical scale. Likewise, λ_n is the actual normalized mean defect, not a coordinate that may be dropped prematurely.

6 The universal cutoff profile and the whole large-action cloud

For $s > 0$ define

$$\Psi_{\alpha,s}(t) = J_\alpha(t) - \int_s^\infty e^{itx} x^{-1-\alpha} dx, \quad (6.1)$$

$$H_{\alpha,s}(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity + \Psi_{\alpha,s}(t)} dt, \quad \Phi_\alpha(s, y) = \frac{H_{\alpha,s}(y)}{g_\alpha(y)}. \quad (6.2)$$

The function H is a subprobability density. The uncompensated integral removed in (6.1) is essential: replacing it by a compensated integral would incorrectly renormalize the truncated model.

An equivalent useful expression is

$$H_{\alpha,s}(y) = e^{-s^{-\alpha}/\alpha} g_{\alpha,s} \left(y + \frac{s^{1-\alpha}}{\alpha-1} \right), \quad (6.3)$$

where $g_{\alpha,s}$ is the probability density with characteristic exponent $\int_0^s (e^{itx} - 1 - itx) x^{-1-\alpha} dx$. This separates the probability of no large jump from the mean shift caused by removing those jumps.

Editorial note (ProveIt, 2026-09-29). In the notation of the critical Hahn article [1], $b_n^{\text{CHT}} = \Gamma(-\alpha)^{1/\alpha} b_n$, $L = s \Gamma(-\alpha)^{-1/\alpha}$, $\lambda^{\text{CHT}} = \Gamma(-\alpha)^{-1/\alpha} \lambda_n$, and $\Phi_\alpha(s, y)$ is its profile R_α at those arguments. With $\ell \equiv 1$, [theorems 6.1](#) and [6.3](#) are its `thm:cutoff` and `thm:budget` (checked at $\alpha = 3/2$, $s = 0.99975$: both give 0.357773962623, the value recorded in `verification/cutoffs.csv`).

Theorem 6.1 (Universal coupling–cutoff law). *If $s_n = M_n/b_n$ remains in a compact subset of $(0, \infty)$, then*

$$\boxed{\frac{u_{n,M_n}(\beta_n)}{u_n(\beta_n)} = \Phi_\alpha(s_n, \lambda_n) + o(1),} \quad (6.4)$$

uniformly on compact sets of (s_n, λ_n) . No slowly varying factor appears in the profile itself; all its leading influence is in b_n , β_n , and ρ_{β_n} .

Proof. Let $B_{n,M}$ be the no-jump-above- M event. Its centered Fourier transform is

$$\mathbb{E} \left[e^{it(K_n - n\beta_n d_1)/b_n} \mathbf{1}_{B_{n,M}} \right] = \exp \left\{ \int (e^{itx} - 1 - itx) \nu_n(dx) - \int_{(M/b_n, \infty)} e^{itx} \nu_n(dx) \right\}.$$

The exponent converges to (6.1). Convergence is uniform for $s = M/b_n$ in compact subsets of $(0, \infty)$: regular variation controls the tails, and the limiting measure has no atom at a moving cutoff. The truncated damping in [theorem 5.1](#) gives L^1 convergence of the Fourier integrands. Consequently

$$b_n \mathbb{P}(K_n = n, B_{n,M_n}) = H_{\alpha,s_n}(\lambda_n) + o(1).$$

Divide by (5.7) and use (2.6). □

Proposition 6.2 (Positivity, a derivative identity, and endpoint values). *For all $s > 0$ and $y \in \mathbb{R}$,*

$$0 < \Phi_\alpha(s, y) < 1, \quad \partial_s \Phi_\alpha(s, y) = s^{-1-\alpha} \frac{H_{\alpha,s}(y-s)}{g_\alpha(y)} > 0. \quad (6.5)$$

Moreover

$$\Phi_\alpha(s, y) \rightarrow 0 \quad (s \downarrow 0), \quad \Phi_\alpha(s, y) \rightarrow 1 \quad (s \rightarrow \infty), \quad (6.6)$$

locally uniformly in y . All these functions are smooth in s, y, α for $s > 0$ and $1 < \alpha < 2$.

Proof. The bounded-jump characteristic function has decay $e^{-c|t|^\alpha}$ for large $|t|$, by the continuum version of the damping proof. Its density therefore extends to an entire function of y . The same is true with half the Lévy intensity. That latter density is nonnegative, not identically zero, and real analytic, so its positive set is open and dense on the real line. The convolution of this density with itself is consequently strictly positive at every real point. This proves $g_{\alpha,s}(y) > 0$. The full stable law is its convolution with the independent compensated large-jump law, so $g_\alpha(y) > 0$ too. Equation (6.3) gives $H > 0$.

The full density decomposes according to the number of jumps above s . Its zero-jump term is $H_{\alpha,s}(y)$; its one-jump term is $\int_s^\infty H_{\alpha,s}(y-x)x^{-1-\alpha}dx$, which is strictly positive. Hence $H < g$. Differentiating (6.1) gives $\partial_s \Psi_{\alpha,s}(t) = e^{its}s^{-1-\alpha}$. Differentiation under the Fourier integral is justified locally by its exponential majorant and gives (6.5).

As $s \rightarrow \infty$, the removed integral in (6.1) has absolute value at most $s^{-\alpha}/\alpha$. Dominated Fourier convergence proves $H \rightarrow g$. For the other endpoint, exponentially tilt the bounded-jump density by $z = c/s$, with a sufficiently small fixed $c > 0$. Its tilted characteristic function has modulus bounded by the untilted one, since its jump intensity is multiplied by $e^{zx} \geq 1$. Splitting its Fourier integral at $1/s$ therefore bounds its density by C/s for $s \leq 1$. With

$$I(c) = \int_0^1 (e^{cv} - 1 - cv)v^{-1-\alpha} dv = O(c^2),$$

(6.3) yields, uniformly for y in a compact set,

$$H_{\alpha,s}(y) \leq \frac{C}{s} \exp \left\{ s^{-\alpha} \left(-\frac{1}{\alpha} - \frac{c}{\alpha-1} + I(c) \right) - \frac{cy}{s} \right\}.$$

The coefficient of $s^{-\alpha}$ is negative for small c . Because $\alpha > 1$, this dominates the $O(s^{-1})$ term and proves $H \rightarrow 0$. Local uniformity and smoothness follow from the same Fourier bounds, including logarithmic factors generated by α derivatives. $\square$

Theorem 6.3 (Sharp action budget). *For any integer sequence $M_n \geq 1$ in a bounded coupling window,*

$$\frac{u_{n,M_n}(\beta_n)}{u_n(\beta_n)} \longrightarrow 1 \quad \Longleftrightarrow \quad \frac{M_n}{b_n} \longrightarrow \infty. \quad (6.7)$$

If $M_n/b_n \rightarrow 0$, the retained fraction tends to zero. If $M_n/b_n \rightarrow s \in (0, \infty)$ and $\lambda_n \rightarrow \lambda$, it tends to $\Phi_\alpha(s, \lambda) \in (0, 1)$.

Proof. The finite positive limit follows from theorem 6.1. If $M_n/b_n \rightarrow \infty$, compare from below with $\lfloor sb_n \rfloor$ for a fixed large s , use monotonicity from theorem 2.1, and then let $s \rightarrow \infty$ in theorem 6.2. The same comparison from above with a fixed small s proves the zero-budget statement. If M_n/b_n does not tend to infinity, a subsequence has bounded ratios; take a further subsequence on which those ratios and λ_n converge. The zero and finite positive cases give a limit strictly below one, a contradiction. $\square$

6.1 A marked point-process formulation

The maximum action is only one observable. Let $\Xi_n = \sum_j N_{n,j} \delta_{j/b_n}$. For a continuous $f \geq 0$ with compact support in $(0, \infty)$, put

$$\Psi_{\alpha,f}(t) = J_\alpha(t) + \int_0^\infty (e^{-f(x)} - 1) e^{itx} x^{-1-\alpha} dx, \quad (6.8)$$

$$H_{\alpha,f}(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity + \Psi_{\alpha,f}(t)} dt. \quad (6.9)$$

Theorem 6.4 (Conditioned large-action cloud). *If $\lambda_n \rightarrow \lambda$, then*

$$\mathbb{E} \left[e^{-\langle f, \Xi_n \rangle} \mid K_n = n \right] \longrightarrow \frac{H_{\alpha, f}(\lambda)}{g_\alpha(\lambda)}. \quad (6.10)$$

The right side is the Laplace functional of the Poisson jump measure ν_α , conditioned on its compensated stable total being λ , with conditioning defined through these continuous densities.

Proof. Poisson marking multiplies the intensity at jump j/b_n by $e^{-f(j/b_n)}$ in the exponential formula, giving exactly the discrete version of (6.8). The extra integral is over a compact set away from zero. All sufficiently small jumps remain unmarked, so the Fourier damping proof is unchanged. Fourier inversion and division by the local limit prove (6.10). To identify the limit, split the Poisson measure at any level below the support of f . Its small-jump component has a strictly positive density; integrating the independent marked large-jump law against that density gives precisely (6.9). This also defines the conditional law at every real λ , not just almost everywhere. Standard Laplace-functional characterization then identifies the locally finite point-process limit away from zero. $\square$

This is a conditioned, not an unconditioned, Poisson cloud. At $1 < \alpha < 2$ the large jumps live on the same scale as the stable total, so conditioning does not become negligible. In particular the retained fraction is generally not the elementary no-jump factor $e^{-s^{-\alpha}/\alpha}$.

7 Exact logarithmic tails and a convergent Lambert chart

From this point until [section 10](#), strengthen the tail assumption to

$$A(t) = a(-\partial_\alpha)^m \text{Li}_{1+\alpha}(t) + P(t), \quad m \in \{1, 2, \dots\}, \quad (7.1)$$

where P is a real polynomial with zero constant term, and the resulting a_j are nonnegative with $a_1 > 0$. Thus the eventual tail is exactly $aj^{-1-\alpha}(\log j)^m$. The polynomial changes any finite prefix. Its coefficients need not themselves be nonnegative.

Define

$$d_k = a(-1)^m \zeta^{(m)}(1 + \alpha - k) + \sum_j [t^j] P(t) j^k. \quad (7.2)$$

For $k = 0, 1$ these agree with (2.2). For $k \geq 2$ they are analytic expansion coefficients, not convergent moments. Parameter differentiation of the convergent polylogarithm identity [5, Eq. 25.12.12] gives, for $|x| < 2\pi$ on a slit branch,

$$A(e^{-x}) = d_0 - d_1 x + a\Gamma(-\alpha)x^\alpha Q_m(\log(1/x)) + \sum_{k \geq 2} \frac{d_k (-x)^k}{k!}, \quad (7.3)$$

where

$$Q_m(L) = \sum_{r=0}^m p_r L^{m-r}, \quad p_r = \binom{m}{r} \frac{\Gamma^{(r)}(-\alpha)}{\Gamma(-\alpha)}, \quad p_0 = 1. \quad (7.4)$$

For example, $p_1 = m\psi(-\alpha)$ and $p_2 = \binom{m}{2}(\psi(-\alpha)^2 + \psi_1(-\alpha))$. The sign of p_1 matters; the differentiation operator is $-\partial_\alpha$, not ∂_α .

At $\beta = \beta_c$, put $C = \beta_c a \Gamma(-\alpha)$ and

$$H(x) = \beta_c \sum_{k \geq 2} \frac{d_k (-1)^k x^{k-2}}{k!}.$$

The exact critical normal form is

$$\varepsilon = Cx^\alpha Q_m(\log(1/x)) + x^2 H(x). \quad (7.5)$$

This is an identity of analytic functions on the slit neighborhood, not merely an asymptotic expansion.

7.1 A Lambert core that keeps the leading logarithm exact

Let $T = e^{-L}$ be the small positive solution of

$$T^\alpha L^m = \varepsilon/C, \quad L > m/\alpha. \quad (7.6)$$

Its exact expression is

$$L = -\frac{m}{\alpha} W_{-1} \left(-\frac{\alpha}{m} (\varepsilon/C)^{1/m} \right), \quad T = e^{-L}. \quad (7.7)$$

The W_{-1} branch is forced by $L \rightarrow \infty$ as $\varepsilon \rightarrow 0$. The other real branch would not describe the critical boundary.

Theorem 7.1 (Absolutely convergent three-coordinate inverse chart). *There is a holomorphic function $V(u, T, R)$ near $(0, 0, 0)$ with $V(0, 0, 0) = 1$ such that the positive critical inverse, and its local complex continuation on compatible branches, is exactly*

$$x = TV \left(\frac{1}{L}, T, \frac{T^{2-\alpha}}{L^m} \right). \quad (7.8)$$

It is characterized by

$$V^\alpha \sum_{r=0}^m p_r u^r (1 - u \log V)^{m-r} + \frac{R}{C} V^2 H(TV) = 1. \quad (7.9)$$

Its Taylor series converges absolutely on a sufficiently small polydisc. After substitution, its monomials are contained in the finite-generator scale

$$\left\{ T^{1+j+(2-\alpha)k} L^{-i-mk} : i, j, k \in \mathbb{N}_0 \right\}. \quad (7.10)$$

There is no finite Puiseux chart in ε for $m \geq 1$.

Proof. Substitute $x = TV$ into (7.5) and divide by $CT^\alpha L^m$. Since $\log(1/x) = L - \log V$, the result is (7.9) after setting $u = 1/L$ and $R = T^{2-\alpha} L^{-m}$. At $(u, T, R, V) = (0, 0, 0, 1)$ the left side minus one vanishes and its derivative in V is $\alpha \neq 0$. The holomorphic implicit-function theorem yields V . Its Taylor series has the usual absolute convergence on smaller polydiscs, and substitution yields (7.10).

These are a finite set of small generators over the exact Lambert core. At a fixed power of T , there may be an entire convergent inverse-logarithmic block; the claim is not local finiteness with respect to the T exponent alone. In the lexicographically ordered power-log scale the support is well based. If α is irrational, the positive T -power generators still have only finitely many pairs (j, k) below any fixed bound.

Finally, $x \sim T$ and $T = \varepsilon^{1/\alpha}$ times a nonconstant logarithmic factor tending to zero. Thus x is not asymptotic to a constant times a rational power of ε , excluding a finite Puiseux chart even when α is rational. $\square$

Proposition 7.2 (First chart coefficients). *With $p_2 = 0$ for $m = 1$,*

$$V(u, 0, 0) = 1 - \frac{p_1}{\alpha}u + \left(\frac{(\alpha+1)p_1^2}{2\alpha^2} - \frac{mp_1}{\alpha^2} - \frac{p_2}{\alpha} \right)u^2 + O(u^3), \quad (7.11)$$

$$[R]V(0, 0, R) = -\frac{H(0)}{\alpha C}. \quad (7.12)$$

Proof. Write $V = 1 + v_1u + v_2u^2 + \dots$. Up to order u^2 ,

$$\sum_{r=0}^m p_r u^r (1 - u \log V)^{m-r} = 1 + p_1 u + (p_2 - m v_1) u^2 + O(u^3).$$

Multiplying by the binomial expansion of V^α and setting the first two coefficients equal to zero gives (7.11). Differentiation of (7.9) in R at the origin gives (7.12). $\square$

7.2 A genuine geometric error bound

Suppose V is bounded by K_0 on a polydisc with radii u_0, T_0, R_0 , and let $V_{\leq N}$ be its Taylor polynomial of total degree at most N . At the physical coordinates put

$$\theta = \max \left\{ \frac{1}{|L|u_0}, \frac{|T|}{T_0}, \frac{|T|^{2-\alpha}|L|^{-m}}{R_0} \right\} < 1.$$

Cauchy's estimate gives

$$|x - TV_{\leq N}(L^{-1}, T, T^{2-\alpha}L^{-m})| \leq |T|K_0 \binom{N+3}{2} \frac{\theta^{N+1}}{(1-\theta)^3}. \quad (7.13)$$

Indeed, degree k contains $\binom{k+2}{2}$ three-variable monomials; summing their geometric majorant proves the bound.

The radii can be chosen by a finite analytic inequality. On $|v - 1| = 1/4$, let $m_\alpha = \min |v^\alpha - 1| > 0$, with powers and logarithms on their branches near one. Choose T_0 so $|Tv|$ remains inside a disc of analyticity of H , and choose u_0, R_0 so the difference between the left side of (7.9) and v^α has modulus less than $m_\alpha/2$ on that circle. Rouché's theorem gives a unique simple zero inside it, uniformly in the parameters. Slightly shrinking the radii then makes $K_0 = 5/4$ a valid Cauchy bound. Numerical evaluation with directed rounding would turn these inequalities into explicit interval certificates; the supplied diagnostic program does not claim to do that step.

8 All-order logarithmic coefficient and cutoff expansions

For the exact logarithmic model, choose the scale by the exact equation

$$b_n^\alpha = n\beta_n a (\log b_n)^m, \quad b_n > e^{m/\alpha}, \quad L_n = \log b_n. \quad (8.1)$$

The large solution is unique for sufficiently large n and is

$$L_n = -\frac{m}{\alpha} W_{-1} \left(-\frac{\alpha}{m} (n\beta_n a)^{-1/m} \right). \quad (8.2)$$

In particular

$$b_n \sim (n\beta_n a)^{1/\alpha} \left(\frac{\log(n\beta_n a)}{\alpha} \right)^{m/\alpha}. \quad (8.3)$$

Replacing (8.1) by (8.3) is valid at leading order but generally invalid for the correction formulas below.

For either $Q_\alpha(t) = J_\alpha(t)$ or $Q_\alpha(t) = \Psi_{\alpha,s}(t)$, define

$$B_k[Q](t) = \sum_{\substack{r_1, \dots, r_m \geq 0 \\ \sum_{h=1}^m h r_h = k}} \prod_{h=1}^m \frac{\left(\binom{m}{h} (-\partial_\alpha)^h Q_\alpha(t) \right)^{r_h}}{r_h!}, \quad B_0[Q] = 1. \quad (8.4)$$

Derivatives in this section always hold t , s , and the normalized coordinate y fixed. They do not differentiate b_n , β_n , d_1 , or λ_n . Write

$$D_k(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity + J_\alpha(t)} B_k[J](t) dt, \quad (8.5)$$

$$N_k(s, y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity + \Psi_{\alpha,s}(t)} B_k[\Psi_s](t) dt. \quad (8.6)$$

The letter D_k here denotes a correction function, not the analytic coefficient d_k in (7.2). In particular $D_0 = g_\alpha$ and $N_0 = H_{\alpha,s}$.

Theorem 8.1 (All fixed orders). *For every fixed nonnegative integer K , the exact logarithmic model has*

$$nb_n \rho_{\beta_n}^n u_n(\beta_n) = \sum_{k=0}^K \frac{D_k(\lambda_n)}{L_n^k} + O(L_n^{-K-1}), \quad (8.7)$$

$$nb_n \rho_{\beta_n}^n u_{n,M_n}(\beta_n) = \sum_{k=0}^K \frac{N_k(s_n, \lambda_n)}{L_n^k} + O(L_n^{-K-1}), \quad (8.8)$$

uniformly for $s_n = M_n/b_n$ and λ_n in compact subsets of $(0, \infty)$ and $\mathbb{R}$, respectively. Finite-prefix changes enter the normalization and algebraically smaller corrections but not the displayed coefficient functions.

Proof. We give the error analysis, since a fixed-frequency formal expansion alone would not prove the theorem. Parameter differentiation of (7.3), with $x = -it/b_n$, gives the centered exponent

$$\log \phi_n(t) = J_\alpha(t) + \sum_{h=1}^m \binom{m}{h} L_n^{-h} (-\partial_\alpha)^h J_\alpha(t) + E_n(t), \quad (8.9)$$

where, for $|t| \leq \theta_0 b_n$,

$$|E_n(t)| \leq C \frac{n}{b_n^2} |t|^2. \quad (8.10)$$

To see the powers of L_n , note that the singular part is $(-\partial_\alpha)^m \{b_n^{-\alpha} J_\alpha(t)\}$ with b_n held fixed during this algebraic differentiation; Leibniz's rule and (8.1) give (8.9). The analytic remainder follows from the convergent analytic series in (7.3). Since $\alpha < 2$,

$$\frac{n}{b_n^2} = \frac{b_n^{\alpha-2}}{\beta_n a L_n^m} = o(L_n^{-r}) \quad \text{for every fixed } r.$$

For the cutoff exponent, subtract $n\beta_n \sum_{j > M_n} a_j e^{itj/b_n}$. For large n all indices here are beyond the finite prefix. The normalized weights on the mesh j/b_n are exactly

$$\frac{1}{b_n} (j/b_n)^{-1-\alpha} \left(1 + \frac{\log(j/b_n)}{L_n} \right)^m.$$

On $s_n \geq s_- > 0$, comparison with its integral over (s_n, ∞) gives an error $O((1 + |t|)/b_n)$. This follows by bounding the integral of the absolute derivative of $e^{itx}x^{-1-\alpha}(1 + \log x/L_n)^m$; its logarithmic moments at infinity are finite, uniformly for $L_n \geq 1$. Expanding the finite polynomial in L_n^{-1} shows that (8.9) becomes

$$\Psi_{\alpha, s_n}(t) + \sum_{h=1}^m \binom{m}{h} L_n^{-h} (-\partial_\alpha)^h \Psi_{\alpha, s_n}(t) + O\left(\frac{n}{b_n^2} t^2 + \frac{1 + |t|}{b_n}\right). \quad (8.11)$$

The identity $(-\partial_\alpha)^h x^{-1-\alpha} = (\log x)^h x^{-1-\alpha}$ proves that the same differential operator acts on both the retained and removed parts, including the killing and centering terms.

Choose a small $\kappa > 0$, for example $\kappa < 1/(4\alpha)$, and split the Fourier integral at $|t| = L_n^\kappa$. Beyond this interval, [theorem 5.1](#) bounds the actual Fourier tails by $\exp(-cL_n^{\kappa(\alpha-\eta)})$, smaller than every inverse power of L_n . The tails of every proposed coefficient integral have the same property. On $1 \leq |t| \leq L_n^\kappa$, the perturbation in (8.9) is $o(1)$ relative to the magnitude of the negative real part of $J_\alpha(t)$. For the truncated exponent, the difference from $J_\alpha(t)$ and its fixed α derivatives is bounded uniformly in s on compact sets; the same conclusion holds for sufficiently large $|t|$. On $|t| \leq 1$, the quantities $|t|^\alpha(1 + |\log |t||^m)$ are bounded, so the perturbations are $O(1/L_n)$ uniformly. The analytic errors in (8.10) and (8.11) are smaller than every inverse-logarithmic order on the retained frequency interval.

Taylor expansion of the exponential through total order K now has an integrable remainder bounded by $C_K L_n^{-K-1}$ times a polynomial in $|t|$ and $|\log |t||$, multiplied by $e^{-c|t|^\alpha}$ outside a fixed interval. Its coefficients are exactly (8.4). Integration proves (8.7) and (8.8). The phase $e^{-it\lambda_n}$ has unit modulus, so compact uniformity in λ_n presents no further error. This also explains why finite prefixes contribute only below all fixed inverse-logarithmic orders once their effect on d_0, d_1, β_c has been retained. $\square$

Remark 8.2 (Uniformity in the exponent). The same argument is uniform for α in any fixed compact subinterval of $(1, 2)$, for a correspondingly controlled family of prefixes. It is not uniform as α approaches either endpoint. The factors controlling small jumps, large compensated jumps, and n/b_n^2 degenerate there.

8.1 The effective-index law

Since $B_1[Q] = -m\partial_\alpha Q$, differentiation under the Fourier integral gives

$$D_1(y) = -m\partial_\alpha g_\alpha(y), \quad N_1(s, y) = -m\partial_\alpha H_{\alpha, s}(y). \quad (8.12)$$

Dividing the two expansions yields the next result.

Corollary 8.3 (A common first correction). *For the exact logarithmic model,*

$$nb_n \rho_{\beta_n}^n u_n(\beta_n) = g_\alpha(\lambda_n) - \frac{m}{L_n} \partial_\alpha g_\alpha(\lambda_n) + O(L_n^{-2}), \quad (8.13)$$

$$\frac{u_{n, M_n}(\beta_n)}{u_n(\beta_n)} = \Phi_\alpha(s_n, \lambda_n) - \frac{m}{L_n} \partial_\alpha \Phi_\alpha(s_n, \lambda_n) + O(L_n^{-2}). \quad (8.14)$$

Equivalently the two leading profiles may be evaluated at

$$\alpha_{\text{eff}, n} = \alpha - \frac{m}{\log b_n}, \quad (8.15)$$

with an $O((\log b_n)^{-2})$ error, provided b_n, s_n , and λ_n are held fixed in this evaluation.

Proof. The quotient expansion gives

$$\frac{-m\partial_\alpha H}{g} - \frac{H(-m\partial_\alpha g)}{g^2} = -m\partial_\alpha(H/g).$$

Taylor's theorem in α proves the last interpretation. $\square$

The effective index has a direct interpretation at the intensity level:

$$x^{-1-\alpha} \left(1 + \frac{\log x}{L_n}\right)^m = x^{-1-\alpha} \left(1 + \frac{m}{L_n} \log x + O(L_n^{-2} \log^2 x)\right).$$

Its first perturbation is the same as that of $x^{-1-(\alpha-m/L_n)}$. This explanation is not a substitute for the Fourier proof: the latter handles the nonuniform logarithms near zero and infinity, the lattice, and the coefficient conditioning.

8.2 Higher orders are not just an index shift

At the next order,

$$B_2[Q] = \binom{m}{2} \partial_\alpha^2 Q + \frac{m^2}{2} (\partial_\alpha Q)^2. \quad (8.16)$$

For example,

$$D_2 = \frac{m^2}{2} \partial_\alpha^2 g_\alpha - \frac{m}{2} \mathcal{F}^{-1}[e^{J_\alpha} \partial_\alpha^2 J_\alpha].$$

The second term would be missed by blindly replacing α everywhere by $\alpha - m/L_n$. For arbitrary order, the quotient coefficients R_k of the retained fraction are computed by the finite recursion

$$R_0 = \Phi_\alpha, \quad R_k(s, y) = \frac{N_k(s, y) - \sum_{j=1}^k D_j(y) R_{k-j}(s, y)}{g_\alpha(y)}. \quad (8.17)$$

Thus $\sum_{k \leq K} R_k L_n^{-k}$ has the same fixed-order uniform error as in [theorem 8.1](#). No convergence in K is claimed.

9 Two-term minimal action budgets

Fix a desired retained fraction $p \in (0, 1)$. By [theorem 6.2](#), the equation

$$\Phi_\alpha(s_p(\alpha, y), y) = p \quad (9.1)$$

has exactly one solution $s_p(\alpha, y) > 0$. The derivative formula [\(6.5\)](#) and the implicit-function theorem show that s_p is smooth and

$$\partial_\alpha s_p(\alpha, y) = -\frac{\partial_\alpha \Phi_\alpha(s_p(\alpha, y), y)}{\partial_s \Phi_\alpha(s_p(\alpha, y), y)}. \quad (9.2)$$

Let $M_n^*(p)$ be the smallest integer M for which $u_{n,M}(\beta_n)/u_n(\beta_n) \geq p$. It exists because the ratio equals one at $M = n$.

Theorem 9.1 (An explicit two-term asymptotic budget). *In the exact logarithmic model and a bounded coupling window,*

$$\frac{M_n^*(p)}{b_n} = s_p(\alpha, \lambda_n) + \frac{m}{L_n} \frac{\partial_\alpha \Phi_\alpha(s_p(\alpha, \lambda_n), \lambda_n)}{\partial_s \Phi_\alpha(s_p(\alpha, \lambda_n), \lambda_n)} + O(L_n^{-2} + b_n^{-1}) \quad (9.3)$$

$$= s_p\left(\alpha - \frac{m}{L_n}, \lambda_n\right) + O(L_n^{-2} + b_n^{-1}). \quad (9.4)$$

The denominator in (9.3) is explicitly positive by (6.5). The integer ambiguity is at most the indicated $O(1)$ in $M_n^*(p)$.

Proof. For y in a compact set the quantiles stay in a compact subinterval of $(0, \infty)$, by local uniformity of (6.6). On a slightly larger compact rectangle, $\partial_s \Phi_\alpha$ has a positive lower bound. Insert $s = s_p + (m/L_n)(\partial_\alpha \Phi / \partial_s \Phi)$ into (8.14); its order- L_n^{-1} terms cancel. The remaining error is $O(L_n^{-2})$ uniformly. Moving s by a sufficiently large constant times L_n^{-2} brackets the target value by the derivative lower bound. Monotonicity of the actual retained fraction then brackets its smallest integer crossing, and rounding changes s by at most $1/b_n$. Equation (9.2) gives the second form. $\square$

This is an asymptotic selection rule, not an unconditional finite- n certificate. A theorem with an unspecified $O(L_n^{-2})$ constant cannot by itself promise a particular retained fraction at a stated finite n . A proof-producing selector would additionally enclose the Fourier integrals, provide effective constants for the error analysis, and round outward.

10 Reproducible calculations and what they actually verify

10.1 Exact checks

The supplied `verification/verify.py` performs 552 exact assertions. It compares direct feedback recursion with Lagrange extraction for three rational couplings through order 16, checks finite-cutoff monotonicity and agreement through the cutoff, verifies the first two chart coefficients symbolically, and checks a Bell-polynomial coefficient recurrence through degree six. These are finite exact algebra checks. They do not certify an infinite Fourier integral or an asymptotic remainder.

10.2 A logarithmic-tail example

The numerical example is

$$\alpha = \frac{3}{2}, \quad a = 1, \quad m = 1, \quad A(t) = t - \partial_s \text{Li}_s(t)|_{s=5/2}. \quad (10.1)$$

Thus $a_1 = 1$ and $a_j = (\log j)/j^{5/2}$ for $j \geq 2$. The constants are

$$\begin{aligned} d_0 &= 1 - \zeta'(5/2) = 1.38734195032620997271\dots, \\ d_1 &= 1 - \zeta'(3/2) = 4.93223973743110151071\dots, \\ \beta_c &= 0.20274764675588095470\dots \end{aligned}$$

At the critical coupling, $\lambda_n = 0$. The first correction constants are

$$\begin{aligned} D_0(0) &= 0.14026098237322958073\dots, \\ D_1(0) &= -0.22110216101538913083\dots, \\ D_2(0) &= -0.13928691395372795423\dots, \\ D_3(0) &= 0.70330022056722715680\dots, \\ D_4(0) &= 0.48955185500793247479\dots \end{aligned}$$

For $m = 1$ these are computed from

$$D_k(0) = \frac{1}{2\pi k!} \int_{\mathbb{R}} e^{J_\alpha(t)} [J_\alpha(t)(\psi(-\alpha) - \log(-it))]^k dt.$$

n	b_n	Observed $b_n \mathbb{P}(K_n = n)$	Through L_n^{-1}	Through L_n^{-4}
256	31.834	0.082557036	0.076368579	0.085122191
1,024	96.559	0.093477009	0.091881350	0.093702738
4,096	279.735	0.101079624	0.101015630	0.101046244
16,384	788.914	0.106639717	0.107115498	0.106601915
65,536	2185.549	0.110858081	0.111507663	0.110838856

Table 1: Coefficient diagnostics for the logarithmic tail at $\alpha = 3/2$. The common leading approximation is $0.140260982\dots$. These are floating-point diagnostics, not certified enclosures.

The expression at $t = 0$ is interpreted by continuity.

The convergence is logarithmically slow. Additional terms need not improve an approximation monotonically at modest n : the signs and sizes of D_k matter, and the smaller analytic and lattice corrections have not been included in this table. The theorem is a fixed-order asymptotic law, not a finite- n alternation or enveloping theorem.

n	Actual M/b_n	Observed retained fraction	$\Phi_{3/2}$	First correction
1,024	0.994215	0.096370999	0.353610742	0.074240642
4,096	0.997372	0.133121494	0.355987344	0.129040344
16,384	0.998842	0.162779389	0.357092276	0.165295316
65,536	0.999749	0.186518804	0.357773963	0.191326680

Table 2: Cutoff diagnostics with $M = \lfloor b_n \rfloor$ and $\lambda_n = 0$. The first correction is $\Phi_{3/2} - (\log b_n)^{-1} \partial_\alpha \Phi_\alpha|_{\alpha=3/2}$.

The leading profile can be visibly inaccurate even at a large index. The first derivative correction substantially improves the displayed $s \approx 1$ values. At smaller s and modest n , a truncated correction can even be negative, although the true probability never is. Such a negative approximation is evidence that the selected truncation is not useful there, not a violation of the probability identity or a certified error estimate.

10.3 Inverse-chart diagnostics

For (10.1), take L as the independent core coordinate, $T = e^{-L}$, and $\varepsilon = CT^{3/2}L$. Solve the exact normal form using 65-digit arithmetic and compare x/T with (7.11). Here $p_1 = \psi(-3/2)$ and $p_2 = 0$.

L	Exact-equation numerical x/T	Through L^{-1}	Through L^{-2}
8	0.939577755920	0.941403613280	0.940812501724
12	0.960595968884	0.960935742186	0.960673025939
20	0.976477564834	0.976561445312	0.976466867463
35	0.986577920376	0.986606540178	0.986575657615

Table 3: Inverse-chart diagnostics. The displayed second-order approximation does not include the much smaller R -coordinate contribution.

The exact equation is evaluated through a parameter derivative of the polylogarithm, not through a truncated inverse series. This separates the numerical check from the formula it is checking.

10.4 Numerical method and independent cross-checks

The coefficient computations use the exact finite-degree dependence of (2.4). A Fourier transform evaluates $\exp(n\beta_c(A_n(re^{i\theta}) - d_0))$ with $r = e^{-2/n}$ and transform length at least $32n$. The coefficient at degree n , divided by r^n , gives $\mathbb{P}(K_n = n)$ up to numerical quadrature and roundoff errors. For a cutoff the constant remains the full d_0 , in accordance with the joint no-large-jump event. A separate positive recurrence,

$$P_0 = e^{-n\beta_c d_0}, \quad P_k = \frac{n\beta_c}{k} \sum_{j=1}^k j a_j P_{k-j},$$

checks the full coefficient at $n = 256$ using 65-digit arithmetic. The measured relative discrepancy is recorded in `results.json`.

Cutoff densities are evaluated by Fourier quadrature, with a Gauss–Jacobi rule for the compensated small-jump integral. Repeating with 96 and 160 nodes checks numerical stability. The derivative in (8.14) is approximated by a centered difference in α , with normalized s and y held fixed. These calculations use ordinary floating-point arithmetic and are not interval enclosures. All numerical tables are generated from the retained machine-readable outputs; no empirical value is used as a lemma in the proofs.

11 Consequences for a transseries implementation

11.1 Three distinct kinds of representation

An implementation should distinguish the following objects instead of forcing them into a single unchecked datatype. First, an asymptotic-inverse object may specify $R^\leftarrow$ and its regular-variation laws without an explicit monomial series. This is the appropriate representation for arbitrary ℓ in [theorem 3.2](#). Second, a convergent local chart has a finite tuple of small coordinates, an analytic defining equation, Cauchy radii, and a majorant such as (7.13). Third, a coefficient-asymptotic object has correction operators and fixed-order remainder statements but need not have a convergent infinite expansion.

The oscillatory example belongs to the first class. Exact logarithmic tails belong to both the second and third. Keeping these distinctions explicit prevents a formal closure claim from being mistaken for an analytic realization theorem.

11.2 A practical coefficient pipeline

For the logarithmic model, a symbolic front end can compute the polynomial Q_m from gamma derivatives, generate the Taylor jet of (7.9), and obtain B_k by the finite recurrence

$$B_0 = 1, \quad kB_k = \sum_{r=1}^{\min(k,m)} r \binom{m}{r} (-\partial_\alpha)^r Q B_{k-r}.$$

A numerical back end then evaluates D_k and N_k with Fourier integrals. The ratio recurrence (8.17) and the positive derivative (6.5) produce cutoff approximations and well-conditioned quantile inversion on compact rectangles.

The physical scale must be obtained from (8.1) or the exact Lambert formula, not from a rough $n^{1/\alpha}(\log n)^{m/\alpha}$ approximation when subleading accuracy is wanted. Similarly, all finite prefix effects on d_0 and d_1 must be retained before applying the universal normalized profile.

11.3 A realistic formalization boundary

No Lean file is included or claimed to compile. A useful formalization sequence would begin with finite formal-power-series algebra: Lagrange coefficient identities, the cutoff dependence, and the Bell recurrence. The next layer is the analytic implicit-function theorem with an explicit polydisc majorant. Those statements can be expressed without formalizing stable probability laws.

The probabilistic layer needs independent Poisson sums, the exact generating function, and Fourier inversion with an L^1 majorant. Only after those interfaces are available should one assemble the regular-variation limits and the triangular-array local theorem. Finally, a directed-rounding quadrature checker could certify profile values and error bounds. Existing repository results on formal support or finite recurrences are useful ingredients, but none is a substitute for these analytic hypotheses.

12 Further research questions and concrete targets

1. Second-order slow variation beyond exact logarithmic powers. Suppose $\ell(bx)/\ell(b) = 1 + A(b)\log x + o(A(b))$ with a weighted uniform remainder. Determine the weakest domination assumptions under which $\Phi_\alpha - A(b)\partial_\alpha\Phi_\alpha$ remains the correct first cutoff approximation. Pointwise second-order variation alone is not an integrable error estimate; the small- and large-jump tails must be controlled. The logarithmic model proves the target identity in one fully specified class, not in this unrestricted second-order class.

2. Noninteger logarithmic powers and convergent charts. For $a_j = aj^{-1-\alpha}(\log j)^r$ with real noninteger r , the leading regular-variation and cutoff theorems already apply. Determine whether the inverse admits an absolutely convergent finite-coordinate chart, or only an asymptotic chart, under a suitable analytic continuation of the tail. Integer parameter differentiation is no longer an exact finite operation.

3. Effective finite- n error constants. Make the constants in [theorem 8.1](#) explicit on a prescribed compact rectangle. Combine interval evaluation of D_k, N_k with the positive profile derivative to produce a genuinely certified $M_n^*(p)$ enclosure. The current two-term formula is asymptotic and the supplied numerical outputs are not outward-rounded certificates.

4. Expanding coupling windows. Find the largest growing range of $|\lambda_n|$ on which the logarithmic expansion remains relatively accurate, and match it to one-large-jump and exponential-tilt regimes. The uniform absolute local theorem does not supply relative accuracy where $g_\alpha(\lambda_n)$ becomes very small.

5. Joint endpoint and slow-variation limits. Let $\alpha \uparrow 2$ or $\alpha \downarrow 1$ while the logarithmic tail and n vary. Derive a common normalization and an error theorem covering the interaction of the endpoint singularity with $m/\log b_n$. Substitution of an endpoint in [\(8.15\)](#) is not justified by this paper.

Editorial note (ProveIt, 2026-09-29). Slowly varying factors at or near $\alpha = 2$ are also asked about by the logarithmic-endpoint package’s Question 6, the marginal package’s Question 2, the stable–Gaussian package’s Question 3 and the confluent package’s Question 4 [\[8, 9, 11, 10\]](#); none is answered.

6. Large order of the inverse-logarithmic expansion. Determine the growth of D_k, N_k as $k \rightarrow \infty$, the optimal truncation index, and whether a Borel or accelerated summation reconstructs the coefficient law. The convergent inverse chart does not imply convergence of this outer expansion. Its complex saddle structure may encode a second level of transseries information.

7. Oscillatory slow blocks as a symbolic language. Develop a representation that includes $2 + \sin(\log \log y)$ without falsely asserting membership in a nonoscillatory power-log scale. A first target is an algebra of periodic functions of $\log \log y$ with explicit asymptotic inversion rules and error control. Decide which coefficient and cutoff computations remain algorithmically tractable in that enlarged language.

8. Perturbed slopes. Replace the exact slopes j in (2.3) by $j + r_j$. Even $r_j = o(j)$ can exponentially reweight the boundary amplitudes through $e^{r_j U}$, so a naive claim that the present theorem is stable under every sublinear perturbation would be false. Identify a correctly normalized perturbation class preserving the slow tail, then determine the new chart generators and correction operators.

9. Multiple marked clouds and functional limits. The Laplace functional in theorem 6.4 determines the limiting point measure away from zero. Develop quantitative joint laws for ordered actions, small-jump compensation, and the induced tree or allocation structures. Prove metric error bounds rather than only convergence of Laplace functionals, and track their sensitivity to the coupling coordinate.

10. Minimal symbolic and analytic certificates. Implement separate checkers for a coefficient identity, a support claim, a convergent analytic chart, and an asymptotic error bound. A robust ProveIt integration should make the distinction visible in its interface: a verified finite coefficient must not silently stand in for a verified infinite asymptotic statement.

13 Conclusion

The slowly varying extension has a positive and a negative side. The inverse scale and action budget remain rigid after the correct regular-variation normalization: the same stable density and the same conditional cutoff profile apply. But slow variation alone does not force a power-logarithmic inverse representation. The explicit oscillatory example separates those two conclusions.

Exact logarithmic tails occupy a useful intermediate class. They admit a convergent Lambert-based chart and also a computable all-order expansion for large coefficients and action retention. At first order, the common correction is an effective shift of the stable index. The strict derivative formula for the cutoff profile makes that correction actionable as a minimal-budget law. This resolves the fixed-interior-exponent part of the repository's slowly-varying-tail question, while leaving endpoint uniformity, unrestricted analytic realization, and fully certified finite- n computation as explicit further problems.

A Normalization and sign audit

The following checks summarize the convention-sensitive points.

Object	Convention and consequence
Lévy measure	$x^{-1-\alpha} dx$, with no gamma factor. Thus $J_\alpha(t) = \Gamma(-\alpha)(-it)^\alpha$.
Stable scale	$n\beta a\ell(b)/b^\alpha \rightarrow 1$. A scale that absorbs $\Gamma(-\alpha)$ requires a correspondingly rescaled density and cutoff.
Coupling coordinate	$\lambda_n = (n - n\beta_n d_1)/b_n$; positive values are targets above the mean.
No-large-jump exponent	$\Psi_s = J - \int_s^\infty e^{itx} \nu(dx)$. The removed term is not compensated.
Mean shift after removal	$H_s(y) = e^{-s^{-\alpha}/\alpha} g_s(y + s^{1-\alpha}/(\alpha - 1))$. The shift has a plus sign inside g_s .
Logarithmic tail	$(-\partial_\alpha)^m \text{Li}_{1+\alpha}$, giving $(\log j)^m j^{-1-\alpha}$.
First profile correction	$-(m/\log b)\partial_\alpha \Phi_\alpha$ at fixed normalized coordinates.
Quantile correction	$+(m/\log b)(\partial_\alpha \Phi/\partial_s \Phi)$, or equivalently $s_p(\alpha - m/\log b, y)$.
Critical inverse	$\varepsilon = x + \beta_c(A(e^{-x}) - d_0)$; $U = \beta_c d_0 + \varepsilon - x$.
Lambert branch	W_{-1} , since both the inverse-core L and the coefficient-scale $\log b$ tend to $+\infty$.

B Reproduction instructions

The package is self-contained apart from standard LaTeX packages and the Python dependencies listed in `requirements.txt`. From the package root, run

```
python verification/verify.py
python verification/make_tables.py --input rerun/results.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The full verification writes `results.json`, three CSV files, and prints its exact-check count and diagnostics. The table generator turns those retained outputs into the three small TeX table fragments input by the article. A quicker run uses `-quick` and stops the coefficient diagnostics at $n = 4096$; it should not be used to regenerate the full tables without restoring the full data afterward.

The supplied PDF was built from the supplied TeX and table fragments. The exact algebra checks were run; the numerical diagnostics were run; the article’s analytic proofs were not checked by a proof assistant.

References

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